Motor Placement Menu

Where Should the Drive Motors Go?

Graduate Expectation Tags

- **[Purple: Curiosity & Creativity]** We explore different ways to fit motors into the chassis.
- **[Red: Critical Thinking & Problem Solving]** We compare turning, space, attachment access, and repair trade-offs.
- **[Green: Personal Development]** We make responsible choices based on available parts, time, and team skill.
- **[Blue: Effective Communication]** We explain motor placement using sketches, evidence, and trade-offs.

Decision Being Made

Your team is deciding where to place the drive motors. Motor placement affects how the robot turns, how easy it is to control, how much open space remains in the chassis, how easy it is to attach future mechanisms, and how easy the robot is to repair.

The drive motors are not just power sources. Their location shapes the whole robot layout.

Background Sources

Use ideas from:

- **Chassis Shape Menu**
- **Chassis Material Menu**
- **Wheel Selection Menu**
- **Chassis Physics and Space Planning**
- robot driving observations,
- available motors, hubs, gears, shafts, and brackets,
- and your team's chassis sketch.

Why This Decision Matters

Motor placement is one of the hidden decisions that affects everything else.

Motors in the center may make the robot easier to turn, but they take up valuable middle space. Motors near the ends may leave the center open, but the robot may need more room to turn. Motors inside channels may look clean and protected, but they can make it harder to add attachments later.

Your team needs to decide what matters most: turning, space, protection, repair access, future mechanisms, or speed of building.

How to Use This Menu

- Review the motor placement options.
- Check your chassis shape and wheel layout.
- Think about where the battery, electronics, shooter, hopper, and trigger might go.

- Choose a first motor placement plan.
- Sketch the motors on your top-view chassis drawing.
- Record your choice, trade-off, and evidence plan in the portfolio.

Menu Options

Option A: Motors Near the Center

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The drive motors and powered wheels are placed closer to the middle of the robot.

What it helps: This can make the robot easier to turn and may help it rotate in place more smoothly. It can make the robot feel more controllable for a human driver.

Best for: Teams that want easier turning, predictable control, and a robot that can line up without needing a huge turning space.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Motor mounts, compatible wheels, hubs, shafts, brackets, spacers, and space in the middle of the chassis.

Possible trade-offs: Motors in the center occupy valuable middle space. This may interfere with a ball path, hopper, shooter mount, battery, electronics, or future intake.

Evidence we might watch: Does the robot turn smoothly? Can it rotate in place? Is there still enough room for the battery, electronics, and future mechanisms?

Option B: Motors Near the Ends

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The motors and powered wheels are placed closer to the front and/or back ends of the robot.

What it helps: This can leave the center of the robot more open for game pieces, electronics, battery, hopper, or mechanisms.

Best for: Teams that need the middle of the robot open for a ball path, hopper, shooter, or intake.

Ease of use: Medium

Likelihood of success: Medium

Time to implement: Moderate

Material / tooling needs: Motor mounts, wheels, hubs, shafts, brackets, and wiring that can reach the motors safely.

Possible trade-offs: The robot may be harder to turn. It may need more space to rotate, and the driver may need more practice to control it.

Evidence we might watch: Does the robot need too much room to turn? Does it line up accurately? Does the open middle space help with future mechanisms?

Option C: Motors Along the Side Rails

Graduate Expectation Connection: [Green: Personal Development] and [Red: Critical Thinking & Problem Solving]

What it means: Motors are mounted along the left and right sides of the chassis, usually near the wheels but not necessarily inside the channel.

What it helps: This can make motors easier to see, wire, repair, and adjust. It may keep the center more open while still supporting a simple drive layout.

Best for: Teams that want repair access and a clear, understandable drivetrain.

Ease of use: Easy to medium

Likelihood of success: High

Time to implement: Fast to moderate

Material / tooling needs: Side mounting brackets, motor mounts, wheels, hubs, spacers, and wire management.

Possible trade-offs: Motors may stick out or be more exposed to contact. They may reduce space for side attachments.

Evidence we might watch: Can we reach the motors and wires? Are the motors protected enough? Do they block future attachments?

Option D: Motors Inside the Channels

Graduate Expectation Connection: [Purple: Curiosity & Creativity] and [Red: Critical Thinking & Problem Solving]

What it means: The motors are mounted inside U-channel or tucked tightly into the frame. This may use compact motor placement, bevel gears, or special brackets.

What it helps: This can protect motors and create a clean, compact drivetrain. It may keep the outside of the robot clear.

Best for: Teams that want a compact design and have enough time, tools, and support to build carefully.

Ease of use: Hard

Likelihood of success: Medium to risky

Time to implement: Slow

Material / tooling needs: Compatible channel, motor mounts, gears or shafts if needed, careful alignment, hex tools, and possibly mentor support.

Possible trade-offs: Once motors are inside the channel, it can be much harder to attach new parts to that channel. Repairs may also be harder. If the motor or gear alignment is wrong, debugging can take a long time.

Evidence we might watch: Can we still attach parts to the channel? Can we reach screws and wires? Does the drivetrain run smoothly without binding?

Option E: Motors Protected but Accessible

Graduate Expectation Connection: [Red: Critical Thinking & Problem Solving]

What it means: The team places motors where they are partly protected by the frame but still easy enough to reach for repair, wiring, and inspection.

What it helps: This balances durability with repair access. It avoids hiding the motors so deeply that they become difficult to fix.

Best for: Teams that want a practical, reliable course robot.

Ease of use: Medium

Likelihood of success: High

Time to implement: Moderate

Material / tooling needs: Motor mounts, brackets, protective frame members, enough space for tools and wires.

Possible trade-offs: The robot may not look as compact as a hidden-motor design, and motors may still take up useful space.

Evidence we might watch: Can the team fix a loose motor or wire quickly? Are motors protected during normal driving and contact?

Option F: Keep Motor Placement From a Recycled Chassis

Graduate Expectation Connection: [Green: Personal Development] and [Orange: Cultural Understanding & Civic Engagement]

What it means: If the team is using a known or recycled chassis, it keeps the existing motor placement unless there is a clear reason to change it.

What it helps: This saves time and reduces the risk of rebuilding the drivetrain incorrectly.

Best for: Teams that want to spend more time on programming, driving, shooter design, hopper, trigger, testing, and portfolio work.

Ease of use: Easy

Likelihood of success: High if the drivetrain already works

Time to implement: Fast

Material / tooling needs: Existing chassis, inspection checklist, replacement screws or wires if needed.

Possible trade-offs: The existing motor placement may limit chassis shape, open space, turning behavior, or future mechanisms.

Evidence we might watch: Does the recycled drivetrain drive reliably? What space does it leave or block? Would changing it cost more time than it saves?

Comparison Table

Option	Best Feature	Main Risk	Good For
Motors near center	Easier turning and rotation	Uses middle space	Driver control and tight turns
Motors near ends	Opens middle space	Harder turning	Game-piece paths and mechanisms
Motors along side rails	Easy access and clear layout	More exposed	Repairable simple drivetrains
Motors inside channels	Compact and protected	Hard to attach/repair later	Advanced compact builds
Protected but accessible	Balanced practical design	Still uses space	Reliable course robot

Option	Best Feature	Main Risk	Good For
Keep recycled placement	Saves time	Limits design choices	Fast reliable start

Motor Placement Questions

Before finalizing motor placement, answer:

- Where are the powered wheels?
- Can the robot turn in the space available?
- Can the driver control the robot easily?
- Does motor placement block the center of the robot?
- Is there room for the battery and electronics?
- Is there room for the shooter, hopper, trigger, or intake?
- Can we reach motor screws and wires later?
- Are motors protected from normal contact?
- Are wires safe from wheels and moving parts?
- Does this placement fit our chassis shape and materials?

Motor Placement Decision Table

Record your decision in your engineering portfolio.

Decision Area	Our Choice	Why We Chose It	What We Gave Up	Evidence We Will Watch
Motor placement				
Powered wheel location				
Space affected				
Repair access				

Must / Should / Could

Level	Expectation
Must Do	Choose where the drive motors will go and show them on a labeled chassis sketch.
Should Do	Explain how motor placement affects turning, driver control, space, and repair access.
Could Do	Compare two motor placements using turning radius, driver feedback, or available space measurements.

SLD Connection

Main SLD Prompt

Should our drive motors be placed for easier turning, more open space, better protection, or easier repair?

Supporting Questions

- Is turning control more important than open space?
- What future mechanisms might need the middle of the chassis?
- How much repair access do we need?
- Should motors be protected even if they become harder to reach?
- What motor placement best fits our chassis shape and wheel choice?
- When is a clean compact design worth the extra build difficulty?

Portfolio Deliverable

Add the following to your engineering portfolio:

- selected motor placement,
- labeled chassis sketch showing motors and powered wheels,
- explanation of why the team chose this placement,
- trade-off statement,
- repair/access notes,
- and evidence the team will watch during testing.

Reflection

Our motor placement choice is:

____

We chose this because:

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The main trade-off we accepted is:

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The evidence we will watch during testing is:

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